Novelty & Prior-Art Differentiation — Smart Refrigerator Monitor

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Executive Summary

Fourteen patents were reviewed against the Smart Refrigerator Health Monitoring & Energy Optimization System (ESP32-based). None of them disclose the same combination of elements this project implements. The closest four each match one piece of the system — a composite health score, a recovery-time-trend maintenance signal, a door-close recovery timer, or the same core sensor set — but every one of them depends on something this project deliberately does without: a cloud server, a real-time clock, machine learning, compressor telemetry, or direct power measurement.

The novelty claim, in one sentence: this is the only reviewed system that derives a same-day composite health score, a trend-based predictive-maintenance flag, and a software energy estimate entirely on-device, from a fixed sensor set (temperature, humidity, door), with zero cloud dependency, zero real-time clock, and zero machine learning — using a persistent software day-counter in flash storage specifically engineered around the fact that its Wi-Fi Access Point has no internet uplink to sync time from.

How this document is organized: Section 2 gives the keep/remove verdict on all 14 patents at a glance. Section 3 explains why five of them don't apply. Sections 4–6 compare the remaining nine point by point against our system, grouped by how close the match is. Section 7 lays the comparison out as a single matrix. Section 8 states the novelty argument directly, keyed to that matrix.

Patent Review Verdict

One note before the table: two of the uploaded files (`US7908875_B2.pdf` and `US7908875_B2_1.pdf`) are the same patent — treat them as one entry, not two.

Patent	Title (short)	Assignee	Verdict	Recommendation
CN103206835B	Multi-zone fridge, compressor power control	Hefei Wangyuan Electronic Tech	LOW	**Remove**
CN104110937B	Frost-detection & auto-defrost device	Jiangsu Shuanglu Electrical Appliance	MEDIUM	Keep (minor)
CN115790060A	NB-IoT dual-sensor door-opening inference	Zhejiang Univ. Hospital (medical)	MEDIUM	Keep (minor)
CN116680530A	Refrigerator health-index method	Haier	HIGH	**Keep (primary)**
CN119042920B	Food-freshness AI/RFID/camera system	Suzhou City University	LOW	**Remove**
CN205678985U	Mechanical door auto-closer	Zhang Shuyan (individual)	LOW	**Remove**
CN213147063U	Lab-fridge cloud door-switch logger	Hangzhou Yinghu Technology	MEDIUM	Keep (minor)
EP4375600A1	Smart Refrigerator (night mode, app alerts)	LG Electronics	MEDIUM	Keep (minor)
US7908875B2	Industrial refrigeration monitoring system	Seth Smith (individual)	HIGH	**Keep (primary)**
US9441991B2	Fridge log data / recovery-time trend display	Panasonic	HIGH	**Keep (primary)**
US10941980B2	Predictive maintenance via ML/DTW on commercial cases	IBM	LOW	**Remove**

Patent	Title (short)	Assignee	Verdict	Recommendation
US12498156B2	RNN/autoencoder anomaly detection	Accruent	LOW	**Remove**
US12698930B2	Cold-storage fleet monitoring (temp+door+humidity)	Polysense Solutions	HIGH	**Keep (primary)**
US12716634B2	Door-pattern-driven compressor control (SVM)	Electrolux do Brasil	MEDIUM	Keep (minor)

Bottom line: remove **CN103206835B, CN119042920B, CN205678985U, US10941980B2, and US12498156B2** — each sits in a different problem domain (direct compressor power control, food-freshness AI, a mechanical door-closer with no electronics, and cloud/ML fault-diagnosis for commercial fleets) with no meaningful technical overlap with a consumer-grade, offline, sensor-driven monitoring device. Keep the other nine; the four marked HIGH are worth citing and distinguishing carefully, the five marked MEDIUM are worth a shorter mention.

Patents Recommended for Removal

CN103206835B (multi-zone refrigerator with compressor power control) is built around measuring real compressor voltage/current to actively regulate motor power, plus physical air-duct dampers for multi-zone climate control and gas sensors for food ripeness. Our system has no current sensor, never touches the compressor, and has no multi-zone hardware — the only thing in common is "a temperature sensor exists."

CN119042920B (fully automatic fresh-keeping system) is a camera + RFID + cloud deep-learning pipeline that identifies individual food items and predicts their freshness for elderly-care dietary recommendations. This is precisely the "food-freshness prediction" category this project explicitly scoped out from the start — different sensors, different cloud-AI architecture, different problem entirely.

CN205678985U (door auto-closer) is a hardwired transistor/NAND-gate circuit with no microcontroller and no software at all — it physically motors the door shut when a person isn't nearby. It shares the idea that "a door left open is bad," but solves it by actuating hardware, which our monitoring-only system is explicitly designed never to do.

US10941980B2 (IBM) and **US12498156B2** (Accruent) are both cloud-hosted machine-learning anomaly-detection pipelines for commercial/facilities-scale refrigeration fleets — one using DTW/STL defrost-cycle signature matching, the other an RNN/autoencoder trained on setpoint-deviation telemetry. Both require continuous cloud connectivity and a trained model; our system runs entirely on one ESP32 with fixed, deterministic threshold/trend rules and no training data, no network dependency, and no fleet-level comparison.

Closest Prior Art, Part 1

CN116680530A (Haier) — closest match on the *health score* concept

Haier's "refrigerator health analysis" method computes a composite evaluation score from component wear data, after-sales repair records, and physical self-check data (door-seal deformation, cabinet integrity), weighted using a formal pairwise "priority diagram" algorithm, then flags an anomaly whenever a new score falls outside a 3-sigma statistically-derived normal range.

	Haier CN116680530A	Our system
Inputs to the score	Component wear, repair history, physical self-check (data we don't have access to)	Temperature, door behaviour, recovery time, energy estimate — all sensor-derived, same day
Weighting method	Formal priority-diagram pairwise comparison, data-derived	Fixed, disclosed weights: 40 / 20 / 20 / 20
Anomaly threshold	Statistically derived (3-sigma / Laida rule) over arbitrary time periods	Deterministic rule: 3 consecutive valid days, strictly increasing, ≥15% total increase
Time basis	Cumulative/statistical across months or years	Same-day only — the score always answers "how did the fridge do today"
Where it runs	Backend analytics service	Entirely on the ESP32 itself
Hardware disclosed	None — this is a data-processing method, not a device	Full sensor set (DS18B20, DHT22, MC-38, OLED, buzzer) and a self-hosted dashboard

Distinguishing point: Haier's patent is the closest match on the *idea* of a composite weighted health number, but it's a backend statistics method over data this project doesn't collect (repair records, cabinet self-check) and has no embedded implementation, no door/temperature-sensor architecture, and no local dashboard at all.

US9441991B2 (Panasonic) — closest match on the *predictive-maintenance* mechanism

Panasonic's patent tracks inside-temperature recovery time after door closure across years of historical data points, graphs the trend on a phone app, and compares it against other fridges from the same production lot to flag declining performance — computed on a remote management server from logs forwarded over a home gateway and the internet.

	Panasonic US9441991B2	Our system
Core mechanism	Door-close → recovery time → multi-year trend → maintenance flag	Door-close → recovery time → 3-day trend → maintenance flag
Where computed	Remote cloud server, aggregating many users' fridges	On-device, single unit, no network required
Cross-device comparison	Yes — compared against same production-lot failure statistics	None — self-contained
Time reference	Real calendar dates from a networked clock	Software day-counter (elapsed `millis()`, no RTC, no NTP)
Trigger rule	Not quantitatively disclosed — shown as a graph with a text recommendation	Explicit rule: strictly increasing across 3 valid days, ≥15% increase
Other sensors used	Compressor RPM directly	None — no compressor connection of any kind
Display	Mobile app only, fed by the cloud	Local dashboard served by the device itself, no app

Distinguishing point: this is the strongest conceptual match in the whole batch — both use door-close recovery time as the maintenance signal — but Panasonic's version is fundamentally a cloud, multi-tenant, RTC-dependent, compressor-telemetry-assisted

system. Ours is the single-unit, offline, NVS-day-counter, no-compressor-telemetry version of the same underlying idea, with a short, explicit, quantitative trigger rule instead of an open-ended graph.

Closest Prior Art, Part 2

US7908875B2 — closest match on the *door-close → recovery-check* mechanism

This is an industrial/commercial refrigeration monitor: after the door closes, a timer starts, and if the compartment temperature hasn't returned to range within an expected window, refrigerant-line thermocouples (T1/T2) are used to decide whether the cause is a compressor malfunction or simply warm product being loaded, alerting a technician either way.

	US7908875B2	Our system
Domain	Industrial/commercial refrigeration (compressor, condenser, evaporator)	Consumer kitchen fridge, retrofit ESP32 add-on
Sensors	Compartment temp + door sensor + refrigerant-line thermocouples (T1, T2, TC1–TC5)	DS18B20 (compartment) + MC-38 (door) + DHT22 (ambient) — no refrigerant-line sensors
Fault classification	Compares line temperatures to distinguish malfunction vs. warm-product loading	No component-level diagnosis — a single compartment-temperature threshold against a fixed recovery target
Result shape	Binary pass/fail alarm for a single event	Graded GOOD/MODERATE/SLOW per event, plus a multi-day trend for predictive maintenance
Audience	Technician/operator, via generic alarm/monitoring interface	End user, via a local self-hosted dashboard
Energy, health score	Not present	Both present (software estimate + composite score)

Distinguishing point: the door-close-then-check-recovery skeleton is genuinely similar, but this patent diagnoses a single event using refrigerant-line sensors we don't have, with no trend analysis, no health score, no energy model, and no local dashboard — it's a binary industrial alarm, not a graded, trending, user-facing system.

US12698930B2 (Polysense) — closest match on the *sensor set and architecture*

A lab/research freezer fleet-monitoring device: internal temperature + door magnetic switch + ambient temperature + ambient humidity, with an onboard RTC, reporting to a hub/server that stores time-series records and raises alerts against configurable thresholds including door-open duration.

	Polysense US12698930B2	Our system
Sensor set	Internal temp, door reed switch, ambient temp, ambient humidity	Same four categories (DS18B20, MC-38, DHT22 for both humidity and secondary ambient temp)
Time reference	Hardware real-time clock (RTC)	No RTC — software day-counter in NVS flash, built specifically for a no-internet device
Architecture	Battery hub + local/remote server + database, fleet of units	Single ESP32 running its own Wi-Fi Access Point, zero server, zero database
Door-duration alerting	Configurable threshold criteria, combinable	Fixed dual threshold: 30s (buzzer + "long opening" flag) and 60s (dashboard ALERT escalation)
Energy estimate	None	Software duty-cycle kWh/day model
Health score	None — raw max/min temp and door-duration reports only	Composite 0–100 weighted score
Predictive maintenance	None — static threshold alerts only	3-day recovery-time-trend rule
Display	Networked "user computer device" via hub/server	OLED on the device + a dashboard served directly by the device, no app, no server

Distinguishing point: this is the closest match on raw sensor selection and the basic "log temp + door, alert on thresholds, show a dashboard" shape — but it's an RTC-based, server/database-backed fleet tool with no energy model, no health score, and no recovery-time predictive-maintenance logic; every one of those is present in our system and absent here.

Medium-Relevance Prior Art

These five share a narrow technical element with our system but differ enough in mechanism or purpose that they're background citations, not close comparators.

CN104110937B (frost-detection/auto-defrost device) shares the idea of a door-open-duration threshold triggering a system response, but its response is active infrared frost-thickness sensing plus a physical heating element for defrosting, with a camera and cloud notifications — a different problem (frost buildup) solved with different hardware (no equivalent in our design).

CN115790060A (NB-IoT hospital cold-chain monitor) uses the exact same DS18B20 sensor we use, and shares our underlying concern — correctly attributing a temperature rise to a door-opening event rather than a real fault. But it has no physical door switch at all; door-opening is *inferred* from the temperature differential between two separately-placed DS18B20 probes, transmitted over cellular NB-IoT to a hospital server. Our MC-38 reed switch gives direct door-state ground truth, so this inferential logic isn't needed, and our architecture has no cellular/cloud link at all.

CN213147063U (lab-fridge door-switch logger) uses a magnetic reed mechanism functionally similar to our MC-38 and tracks door-open count/duration, but is a thin cloud-IoT wrapper — device → router → WiFi → internet → cloud server → webpage/app — with no temperature sensing, no energy estimate, no health score, and (by the patent's own description) "conventional" internals with no embedded intelligence.

EP4375600A1 (LG Smart Refrigerator) is the most "modern app-connected fridge" of the batch — it shares the general concept of displaying temperature/door/energy state and pushing alerts, but its actual novel features (a sunset/sunrise-synced night mode, ice-maker/filter/water-usage tracking) have no counterpart in our system, and it depends on a home gateway, cloud server, and mobile push notifications — plus an internet-synced clock for the sunset/sunrise feature, which is exactly the dependency our software day-counter exists to avoid.

US12716634B2 (Electrolux) counts door-openings per hour of day — directly analogous to our peak-usage-hour metric — and states an energy-reduction goal, but uses that data to actively drive an SVM machine-learning classifier that changes real compressor and defrost-heater behavior in real time ("shopping mode," "vacation mode"). Our system never actuates anything; it is a passive estimator, and it uses no machine learning anywhere.

Feature Comparison Matrix

The table below lines our system up against the four HIGH-relevance patents across the dimensions that matter for a novelty argument. No single prior-art reference has more than two or three of these at once; our system is the only row with all of them.

Dimension	Our System	Haier CN116680530A	Panasonic US9441991B2	US7908875B2	Polysense US12698930B2
Runs fully offline, no cloud/server	Yes	No (backend service)	No (cloud server)	Partial (local alarm device, but generic network interface)	No (hub + server)
Needs no real-time clock	Yes (software day-counter in NVS)	N/A (no device)	No (networked clock)	N/A (not disclosed)	No (onboard RTC)
Composite 0–100 health score	Yes	Yes (different inputs/method)	No	No	No
Predictive maintenance from recovery-time trend	Yes (3-day, ≥15% rule)	No (uses repair/wear data instead)	Yes (multi-year, no disclosed rule)	No (single-event binary alarm)	No (static threshold alerts only)
Software energy/duty-cycle estimate	Yes	No	No	No	No
Dual door-duration thresholds (buzzer + alert)	Yes (30s / 60s)	No	Not disclosed	No	Configurable but not this scheme
Peak-usage-hour tracking	Yes	No	No	No	Not disclosed

Dimension	Our System	Haier CN116680530A	Panasonic US9441991B2	US7908875B2	Polysense US12698930B2
Local self-hosted dashboard (device-served)	Yes	No (no device at all)	No (mobile app via cloud)	No (generic alarm interface)	No (networked client via server)
Uses machine learning	No	No (statistical, not ML)	Not disclosed	No	No
Direct compressor/mains control	No	No	No	No	No
Direct power/current measurement	No (software estimate)	Not disclosed	Not disclosed	No	No

Reading the matrix: every prior-art reference is missing at least four of these ten properties. The two systems that come closest on the *maintenance* dimension (Panasonic, and to a lesser extent US7908875) are both missing the health score, the energy estimate, and the offline/no-RTC architecture. The one system closest on *sensors and dashboard shape* (Polysense) is missing the health score, the energy estimate, and the predictive-maintenance logic entirely, and depends on an RTC and a server our system has neither of.

Novelty Synthesis & Conclusion

None of the fourteen reviewed patents disclose, individually or in any combination, the specific set of design choices this project implements together:

1. **A composite same-day health score** (40% temperature, 20% door, 20% recovery, 20% energy) computed entirely from live sensor behaviour, with fixed, disclosed weights and thresholds — as opposed to Haier's repair-history/self-check-based statistical method, or no score at all in every other reference.
2. **A deterministic, quantitative predictive-maintenance rule** — three consecutive valid days, strictly increasing average recovery time, ≥15% total increase — as opposed to Panasonic's open-ended multi-year graph with a cloud-side recommendation, or no trend logic at all in the rest.
3. **A software-only energy estimate** built from a duty-cycle model (temperature/door/recovery impact scores feeding a baseline duty cycle and a reference wattage) with no current sensor — a feature absent from every single one of the fourteen patents reviewed.

4. **Zero real-time clock**, replaced by a persistent software day-counter stored in flash (Preferences/NVS), specifically engineered around the fact that the device's own Wi-Fi Access Point has no internet uplink to sync time from — every prior-art reference with any time-tracking at all uses either a hardware RTC or a networked clock.
5. **Fully offline, single-unit, device-served dashboard** — the ESP32 hosts its own Wi-Fi network and serves its own webpage with zero cloud server, router, mobile app, or external dependency — as opposed to every reviewed reference with a dashboard, all of which route through a server, gateway, or hub.
6. **No machine learning anywhere in the pipeline** — every score and flag is a disclosed, auditable arithmetic rule, in contrast to the ML-based approaches (IBM, Accruent, Electrolux) found elsewhere in the same general problem space.
7. **Monitoring only, no actuation** — the system never touches the compressor, a heating element, or mains power, unlike CN205678985U (motorized door-closer), CN104110937B (defrost heater), or US12716634B2 (active compressor control), each of which turns its sensing into a physical control action.

Conclusion: the project's technical contribution is not any single sensor, threshold, or formula in isolation — each of those individually has some precedent somewhere in this patent landscape. The novelty is the specific combination: a fully offline, RTC-free, ML-free, actuation-free embedded device that turns a fixed low-cost sensor set into a same-day composite health score, a software energy estimate, and a quantitative recovery-time-trend maintenance warning, all computed on-device and presented through a dashboard the device serves itself. No reviewed patent discloses that combination.